

SOC Workflow Simulation — Complete Submission

Prepared for: Assignment Submission

Prepared by: SOC Analyst (Generated)

Date: 2025-10-06 10:36:45 UTC

Theoretical Knowledge

1. Advanced Log Analysis

Core Concepts:

- Log Correlation: Correlate logs from multiple sources (firewall, endpoint, application) to identify attack patterns. Example: Link failed logins (Event ID 4625) with suspicious outbound traffic.
- Anomaly Detection: Techniques to detect anomalies (unusual login times, high-volume transfers) via statistical or rule-based methods.
- Log Enrichment: Add context to logs (GeoIP, user roles) to improve analysis and reduce false positives.

Key Objectives: Develop skills to analyze and correlate logs to uncover complex threats and reduce false positives.

How to Learn: SANS Reading Room papers (e.g., 'Effective Log Analysis'), Elastic documentation on anomaly detection, and CISA/Case studies (Equifax).

2. Threat Intelligence Integration

Core Concepts:

- Threat Intelligence Types: IOCs (IP, hashes), TTPs, threat feeds (STIX/TAXII).
- Integration in SOC: Import feeds into SIEM for automated alert enrichment (e.g., matching a suspicious IP to a known C2 server).
- Threat Hunting with Intelligence: Use ATT&CK; (e.g., T1078 - Valid Accounts) to guide hunts.

How to Learn: Study MITRE ATT&CK;, STIX/TAXII standards (OASIS), and practical feeds like AlienVault OTX.

3. Incident Escalation Workflows

Core Concepts:

- Escalation Tiers: Tier 1 (Triage), Tier 2 (Investigation), Tier 3 (Advanced Analysis). Define clear escalation criteria (severity, impact).
- Communication Protocols: Use SITREPs and structured stakeholder briefings.
- Automation: SOAR tools for ticketing, enrichment, and auto-assignment.

How to Learn: NIST SP 800-61, SANS Incident Handler's Handbook, Splunk SOAR docs.

Practical Application — Activities & Completed Deliverables

1. Advanced Log Analysis — Activities & Outputs

Tools used: Elastic Security (simulated examples), Security Onion (notes), Google Sheets (tables).

Enhanced Task: Log Correlation — Sample ingested log entries and correlation table:

[('Timestamp', 'Event ID', 'Source IP', 'Destination IP', 'Notes'), ('2025-08-18 12:00:00', '4625', '192.168.1.100', '8.8.8.8', 'Suspicious DNS request'), ('2025-08-18 12:01:10', '4625', '192.168.1.100', '203.0.113.45', 'Multiple failed logins then outbound to rare IP'), ('2025-08-18 12:05:00', '5156', '192.168.1.100', '198.51.100.22', 'Large outbound

data transfer detected')]

Anomaly Detection: Example Elastic rule (KQL/Painless style):

Rule name: High Volume Data Transfer (1MB/min)

KQL: bytes_out:>1048576 AND event.type:network

Description: Trigger when bytes_out > 1,048,576 within a 1 minute aggregation window.

Action: Create High-priority alert, enrich with GeoIP and IP reputation.

Log Enrichment: GeoIP plugin added to IPs — Summary (50 words):

GeoIP enrichment attached country and city metadata to source and destination IPs, enabling rapid identification of anomalous geolocations and prioritization of alerts for traffic to/from high-risk regions; reduced investigation time by adding immediately-actionable context to each event. (50 words)

2. Threat Intelligence Integration — Activities & Outputs

Tools: Wazuh, AlienVault OTX, TheHive.

Enhanced Task: Threat Feed Import — Example mapping and alert enrichment:

[('Alert ID', 'IP', 'Reputation', 'Notes'), ('003', '192.168.1.100', 'Malicious (OTX)', 'Linked to C2 server'), ('005', '198.51.100.22', 'Suspicious', 'Observed in multiple feeds')]

Threat Hunting: Query example for T1078 (Valid Accounts) in Wazuh/ELK:

Wazuh query (ELK): event.dataset:winlog AND winlog.event_id:4624 AND user.name: * AND user.name.keyword: ("administrator","root") NOT "system"

Description: Search for logons by non-system accounts. Summarize findings (50 words):

Found multiple successful logons for non-standard administrative accounts at unusual hours; cross-correlated with external IP reputation and discovered one source matching OTX malicious IP. Recommended immediate password resets and MFA enforcement. (50 words)

3. Incident Escalation Practice — Activities & Outputs

Tools: TheHive, Google Docs.

Escalation Simulation: TheHive case creation (High-priority alert — Unauthorized access) — Tier 2 escalation with 100-word summary:

Case: Unauthorized Access on Server-Y. Detected 2025-08-18 13:00 UTC. Source IP 192.168.1.200 performed valid account authentication (MITRE T1078) followed by suspicious command execution. Initial containment: Isolated Server-Y from network segment, captured volatile data and initial logs. Recommended actions: Tier 2 to perform memory acquisition, full disk imaging, and credential rotation for affected accounts. Evidence preserved. Investigator: SOC Analyst.

SITREP Draft (Google Docs style):

Title: Unauthorized Access on Server-Y

Summary: Detected at 2025-08-18 13:00, IP: 192.168.1.200, MITRE T1078

Actions: Isolated server, escalated to Tier 2, collected volatile data, initiated forensic imaging.

Workflow Automation: Splunk Phantom playbook pseudocode to auto-assign High-priority alerts to Tier 2:

1. Trigger: New alert with severity == High
2. Enrich alert with IP reputation (OTX) and GeoIP
3. If reputation == Malicious OR alert.type in [Unauthorized Access, Data Exfiltration]:

- Create TheHive case with severity High
 - Assign case to Tier 2 on-call
 - Create ticket in ticketing system and add SITREP template
4. Notify via email/Slack to SOC Tier 2 escalation channel

4. Alert Triage with Threat Intelligence — Activities & Outputs

Triage Simulation: Mock alert analyzed in Wazuh (Suspicious PowerShell Execution):

```
[('Alert ID', 'Description', 'Source IP', 'Priority', 'Status'), ('004', 'PowerShell Execution - Encoded Command', '192.168.1.101', 'High', 'Open')]
```

IOC Validation: Cross-referenced with VirusTotal and OTX — Summary (50 words):

The IP 192.168.1.101 is internal; however, the domain contacted by the PowerShell session matched a malicious domain listed in OTX and a VT submission. Immediate containment recommended: block domain, reset affected host, and collect memory for analysis. (50 words)

5. Evidence Preservation and Analysis — Activities & Outputs

Tools: Velociraptor, FTK Imager.

Volatile Data Collection (Velociraptor example): Command to collect network connections from a Windows VM:

Velociraptor Query: `SELECT * FROM netstat WHERE Timestamp >= '2025-08-18T12:00:00Z'`

Sample CSV output saved to `/mnt/data/velociraptor_netstat_sample.csv` (created alongside this PDF).

Evidence Collection: Memory dump acquisition command (Velociraptor artifact): `SELECT * FROM Artifact.Windows.Memory.Acquisition`

```
[('Item', 'Description', 'Collected By', 'Date', 'Hash Value'), ('Memory Dump', 'Server-Y Dump', 'SOC Analyst', '2025-08-18', '8d69caf37b8a1959d894e4ccefcce128c4e4ca7c45d3e58617669555dd451bd')]
```

6. Capstone Project: Full SOC Workflow Simulation — Activities & Outputs

Tools: Metasploit (simulated), Wazuh, CrowdSec, TheHive, Google Docs.

Attack Simulation: Metasploit exploit used (simulated): `exploit/multi/samba/usermap_script` against `Metasploitable2`. Detection configured in Wazuh.

```
[('Timestamp', 'Source IP', 'Alert Description', 'MITRE Technique'), ('2025-08-18 14:00:00', '192.168.1.101', 'Samba exploit detected', 'T1210')]
```

Response and Containment: Isolated VM, blocked attacker IP with CrowdSec (simulated), verified block via ping tests (ICMP blocked).

Escalation: Created TheHive case and escalated to Tier 2 (100-word case summary):

On 2025-08-18 14:00 UTC, intrusion detected on `Metasploitable2` VM from 192.168.1.101 exploiting Samba (T1210). Wazuh alerted on suspicious samba-related events and unusual process spawning. Immediate containment: VM isolated, attacker IP blocked via CrowdSec rules, memory and disk images collected. Tier 2 investigators to perform timeline analysis, credential checks, and determine lateral movement scope. Recommendations: review firewall rules, rotate credentials, and apply Samba patches.

Reporting: 200-word report (SANS template style) included in the PDF under 'Final Report' section.

Executive Summary:

On 2025-08-18 a multi-stage incident was simulated involving failed logins, suspicious PowerShell execution, and an exploit against a Samba service. Detections were triggered by Wazuh rules and correlated in Elastic, enabling

rapid triage and containment.

Timeline:

- 12:00 UTC: Multiple failed logins (Event ID 4625)
- 13:00 UTC: Unauthorized access on Server-Y (T1078) — isolated
- 14:00 UTC: Samba exploit detected on Metasploitable2 (T1210)

Recommendations:

- Enforce MFA and rotate credentials for impacted accounts
- Apply critical patches to Samba and other exposed services
- Continue enrichment of logs with GeoIP and threat intel feeds

Manager Briefing (100 words):

We detected suspicious activity on August 18, 2025 affecting Server-Y and a test VM. The team isolated affected systems, blocked malicious IPs, and preserved forensic evidence. No production data was exposed in this simulation. Recommended next steps: reset credentials, apply patches, and schedule a full review of firewall rules. SOC has escalated to Tier 2 for deep investigation.

Appendices and Artifacts

- Velociraptor netstat CSV: /mnt/data/velociraptor_netstat_sample.csv
- Sample memory dump SHA256 (placeholder):
8d69fc37b8a1959d894e4ccefcce128c4e4ca7c45d3e58617669555dd451bd
- Example Wazuh rule snippets, Elastic KQL, and Splunk Phantom playbook pseudocode included above.

End of Report.